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2027 CYCLE

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CUET UG

Environmental Science (Code 307)

Development and Environment

Economic Growth Limits, Commons Governance, Urban Morphology, Carrying Capacity & Eco-Equity

INSIDE
Concept Notes

INSIDE
Solved Examples

INSIDE
Practice MCQs + Solutions

Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment & CUET Exam Weightage

Syllabus Component	Core Analytical Dimensions & Thinkers	Expected MCQs	Marks Weightage
Growth Limits	GDP/GNP critique, Herman Daly's Steady-State Economics, Club of Rome's **The Limits to Growth** (1972) World3 model, Ecological Footprint (Rees & Wackernagel), Earth Overshoot Day.	2 - 3 MCQs	10 - 15 Marks
The Commons	Hardin's **Tragedy of the Commons** (1968), CPRs (high rivalry, high exclusion cost), Elinor Ostrom's 8 Design Principles (**Governing the Commons** 1990), Polycentricity.	3 MCQs	15 Marks
Urbanization	Push-pull migration, sprawl, Urban Heat Island (UHI delta 2-8°C), low albedo materials, canyon trapping, runoff coefficient surge, flash floods, slum civic deficits.	2 - 3 MCQs	10 - 15 Marks
Urban Planning	Transit-Oriented Development (TOD 400-800m), BRTS, Sponge Cities, SuDS, native tree buffers, Green Building Ratings: GRIHA (India/TERI), LEED, IGBC, ECBC.	2 - 3 MCQs	10 - 15 Marks
Carrying Capacity	Supportive vs Assimilative capacity, IPAT equation ($I = P \times A \times T$), Rockström's 9 Planetary Boundaries, Brundtland (1987), Inter/Intragenerational equity, Warren County (1982).	2 - 3 MCQs	10 - 15 Marks

CUET UG UNIT 4 STRATEGIC EXAM POINTERS

- **1. Hardin vs. Ostrom:** Hardin assumed **unregulated open access** with zero rules. Ostrom proved that **Common Pool Resources (CPRs)** can be sustained via self-governing institutions and 8 design principles.
- **2. Steady-State Axiom:** Daly established that the economy is an **open subsystem** inside a **finite, closed ecosphere**. Constant stocks of capital and people must be kept at minimal throughput.
- **3. The IPAT Identity:** In $I = P \times A \times T$ (Ehrlich & Holdren), environmental impact equals Population times Affluence times Technology.
- **4. UHI & Runoff:** UHI delta reaches 2°C to 8°C due to low albedo and canyon trapping. Runoff coefficient surges from 10% to over 55%.
- **5. GRIHA Identity:** India's indigenous rating tool is **GRIHA** (TERI/MNRE), scoring 34 criteria on a 100-point scale.

1. Conventional Development Paradigms, Economic Metrics & Growth Limits

1.1 Neoclassical Growth Fallacies and the Critique of GDP

Traditional neoclassical economics posits continuous exponential growth as the primary metric of civilizational progress. Economic welfare is evaluated through macroeconomic aggregates:

1. **Gross Domestic Product (GDP):** The total monetary market value of all finished goods and services produced within a country's borders during a specific period.
2. **Gross National Product (GNP):** The total market value of goods and services produced by residents and enterprises of a nation, regardless of geographic location.

Critical Ecological Pathologies of GDP Accounting:

- **Treatment of Natural Capital as Infinite:** Neoclassical accounting treats raw timber, clean freshwater, marine fisheries, and topsoil fertility as free gifts of nature ("zero cost"). Depletion of non-renewable minerals and drawdown of fossil aquifers are recorded as positive income generation rather than capital depreciation.
- **Perverse Defensive Expenditures:** GDP fails to distinguish between welfare-enhancing economic activities and ecological remediation. For instance, an industrial oil spill generates cleanup contracts, litigation fees, and hospital treatments, all tallied as positive additions to GDP.
- **Externalization of Environmental Costs:** Industrial polluters discharge toxic particulates (PM2.5, SO2), chemical effluents, and greenhouse gases (CO2, CH4) without paying economic restitution, leaving corporate profitability artificially inflated.

1.2 Alternative Welfare Indicators: GPI, ISEW and HDI

To rectify the biophysical blindness of GDP, ecological economists formulated alternative welfare metrics:

- **Genuine Progress Indicator (GPI) & Index of Sustainable Economic Welfare (ISEW):** Developed by Herman Daly, John Cobb, and Clifford Cobb. GPI starts with personal consumption, adds non-monetary contributions (volunteer labor, parenting), and systematically subtracts:
  - Economic costs of environmental degradation (wetland loss, farmland soil erosion, air/water pollution damage).
  - Depletion of non-renewable fossil energy assets and mineral stocks.
  - Societal costs including crime, divorce, income inequality, and long-term carbon emissions.
  - **The Threshold Hypothesis:** Manfred Max-Neef noted that while GDP increases, GPI initially rises alongside it but eventually decouples and begins to decline once hidden social and ecological costs exceed growth benefits.
- **Human Development Index (HDI):** Formulated in 1990 by Mahbub ul Haq and Amartya Sen under the UNDP. Evaluates development across three dimensions: **Health** (life expectancy), **Education** (mean/expected years of schooling), and **Standard of Living** (GNI per capita in PPP).

1.3 Herman Daly's Steady-State Economics (SSE)

American ecological economist **Herman Daly** (1938–2022), grounded in Nicholas Georgescu-Roegen's entropy laws, formulated **Steady-State Economics**:

1. **Thermodynamic Subsystem Axiom:** The human economy is an **open thermodynamic subsystem** embedded within a **finite, non-growing, closed biophysical ecosphere**. It takes in low-entropy matter and energy from source sinks, metabolizes them, and discharges high-entropy waste and heat into absorption sinks.
2. **Definition:** An economy with constant physical stocks of artifacts (capital) and people (population), maintained by a **minimal rate of physical throughput**. Birth rates equal death rates at low mortality/fertility levels; production equals depreciation at low rates.
3. **Daly's Operational Principles:**
   - **Renewables:** Harvest rates must not exceed natural regeneration rates (e.g. sustained timber harvest).
   - **Non-Renewables:** Depletion rates must not exceed the development rate of renewable substitutes (e.g. solar replacing coal).
   - **Waste Emission:** Discharges must not exceed the assimilative and recycling capacity of the surrounding environment.

1.4 The Limits to Growth (Club of Rome, 1972)

In 1972, the **Club of Rome** commissioned an MIT team led by **Donella Meadows, Dennis Meadows, Jørgen Randers, and William Behrens III** to model biophysical limits:

- **The World3 Model:** System dynamics computer simulation modeling five global subsystems: (1) Industrial capital per capita, (2) Population, (3) Food production, (4) Non-renewable resource depletion, and (5) Persistent pollution accumulation.
- **Key Findings:** Under the **Standard Run (Business-as-Usual)**, exponential economic and demographic growth collides with biophysical limits, triggering involuntary **overshoot and collapse** during the first half of the 21st century (2020–2050).
- **Equilibrium Solution:** Planetary collapse can only be averted by establishing demographic and economic equilibrium (stabilizing throughput while restoring soils and recycling materials).

1.5 Ecological Footprint and Earth Overshoot Day

In the 1990s, **William Rees** and **Mathis Wackernagel** introduced the **Ecological Footprint (EF)** accounting framework:

- **Definition:** Total area of biologically productive land and water (in **global hectares, gha**) required to produce consumed resources and assimilate generated wastes.
- **Biocapacity (BC):** Capacity of an ecosystem to regenerate biological resources and absorb waste gases.
- **Balance Sheet: Ecological Deficit** occurs when EF $>$ BC; **Ecological Reserve** occurs when BC $>$ EF.
- **Earth Overshoot Day (EOD):** Calculated annually by the **Global Footprint Network**:

$$\text{Earth Overshoot Day} = \left(\frac{\text{Planet's Biocapacity}}{\text{Humanity's Ecological Footprint}} \right) \times 365$$

  - In 1970, EOD fell on December 30. Today, it falls in early August, meaning humanity consumes at a rate of **1.7 Earths**.

1.6 Kenneth Boulding's "Spaceship Earth" & Max-Neef's Human Scale Development

In his seminal 1966 essay, *"The Economics of the Coming Spaceship Earth"*, economist **Kenneth Boulding** contrasted two civilizational worldviews:

- **The Cowboy Economy:** Characteristic of open frontiers, reckless resource extraction, and unrestrained waste dumping. Success is measured by the volume of throughput from production to consumption.
- **The Spaceman Economy:** Acknowledges the Earth as a single spaceship, devoid of unlimited reservoirs for extraction or pollution sinks. Throughput must be minimized, and ecological health is measured by the nature, quality, and complexity of total capital stock rather than extraction velocity.
- **Manfred Max-Neef's Human Scale Development:** Chilean economist Manfred Max-Neef posited that human needs are finite, universal, and classifiable into nine fundamental categories (Subsistence, Protection, Affection, Understanding, Participation, Idleness, Creation, Identity, and Freedom). Wealth accumulation beyond vital needs does not elevate human well-being, confirming the decoupling of GPI from GDP.

2. Resource Depletion, Common Pool Resources & Ostrom's Polycentric Governance

2.1 Natural Resource Classification & Exploitation Dynamics

1. **Non-Renewable Resources:** Finite stock deposits formed over geological timescales (coal, oil, gas, metallic ores, phosphate rock). Once extracted, they cannot regenerate on a human temporal scale.
2. **Renewable Resources:** Flow resources that naturally replenish (solar, wind, timber, fish stocks). They remain sustainable only if harvesting stays strictly below the **Maximum Sustainable Yield (MSY)**. Beyond this threshold, reproductive collapse transforms them into non-renewable stocks.

2.2 Garrett Hardin's "Tragedy of the Commons" (1968)

In December 1968, **Garrett Hardin** published *"The Tragedy of the Commons"* in **Science**:

- **The Pasture Parable:** Rational herdsmen share an open pasture. Each asks: **"What is the utility to me of adding one more animal?"**
- **Mathematical Utility Asymmetry:** The herdsman captures the full positive utility (+1) from selling the animal, but the negative utility of overgrazing (soil erosion, pasture depletion) is shared among all n herdsmen ($-\frac{1}{n}$).
- **The Trajectory:** Pursuing individual self-interest inexorably leads to collective ruin: **"Freedom in a commons brings ruin to all."**
- **Hardin's Prescribed Remedies:** (1) Total privatization (fencing the commons), or (2) Coercive state regulation ("mutual coercion, mutually agreed upon").

Urbanization Pressures, Spatial Morphology & Sustainable City Diagnostics

Comparative Matrix: Demographic Drivers, Urban Heat Island, Hydrology, Green Building Codes & Invariants

CUET UG · EVS UNIT 4 · FIG 1

Urban Vector	Ecological Stressor	Biophysical Mechanism	Sustainable Planning Response	CUET Core Invariant
Spatial Land Use	Urban Sprawl & Fragmentation	Pervious agricultural soil replaced by concrete; per capita land consumption surges.	Compact city design, urban growth boundaries (UGB), infill redevelopment.	Sprawl accelerates habitat fragmentation and motor vehicle dependency.
Microclimate	Urban Heat Island (UHI)	Low albedo asphalt/concrete absorbs solar flux; tall street canyons trap longwave IR radiation.	High-albedo cool roofs, green vegetative roofs, urban shade tree canopy corridors.	UHI temperature delta (ΔT) reaches 2°C to 8°C higher than surrounding rural plains.
Urban Hydrology	Flash Floods & Depletion	Impervious surface ratio surges; stormwater runoff fraction jumps from ~10% to over 55%.	Sponge City design, Sustainable Drainage Systems (SuDS), rooftop rainwater harvesting.	Runoff peaks dramatically; baseflow and groundwater recharge sharply collapse.
Transport & Air	Street Canyon Smog & Noise	Concentrated tailpipe combustion (PM2.5, NOx, CO); high building height-to-width ratio limits dispersion.	Transit-Oriented Development (TOD), Bus Rapid Transit (BRTS), non-motorized lanes.	TOD concentrates high-density zoning within 400-800m of mass rapid transit hubs.
Built Environment	High Embodied Carbon	Extensive energy and water footprints in HVAC, cement, glass, and artificial lighting.	GRIHA (TERI/MNRE rating), LEED (USGBC), IGBC rating, ECBC building code.	GRIHA is India's indigenous green building rating tool evaluating lifecycle energy.
Social Ecology	Environmental Injustice	Disproportionate siting of municipal landfills, waste incinerators, and polluting zones near slums.	Inclusive decentralized civic infrastructure, community tenure, participatory zoning.	Environmental racism / injustice identified in Warren County 1982 PCB protests.

OURMOCKS DEVELOPMENT & ENVIRONMENT MATRIX CODE 307 · UNIT 4: DEVELOPMENT AND ENVIRONMENT PAGE 5 CONCEPT MATRIX

Figure 1: Comparative Matrix: Urbanization Vectors, Thermal Microclimates, Impervious Surfaces, and Green Building Rating Systems (CUET Unit 4).

2.3 The Economic Taxonomy of Goods

Institutional economics classifies goods along two independent axes:

1. **Subtractability (Rivalry):** Does one user's consumption diminish availability for others?
2. **Excludability:** How costly is it to prevent potential beneficiaries from consuming the good?

Excludability \ Subtractability	High Subtractability (Rival)	Low Subtractability (Non-Rival)
High Cost of Exclusion (Difficult)	**Common-Pool Resources (CPRs):** Aquifers, pastures, fisheries, irrigation canals.	**Public Goods:** Climate stability, ozone layer, national defense.

, [**Low Cost of Exclusion (Easy)**], [**Private Goods:**
Farmland, commercial timber, personal vehicles.], [**Club / Toll Goods:**
Toll highways, piped water networks, subscriptions.])

2.4 Elinor Ostrom's Nobel Revolution: Governing the Commons (1990)

Political economist **Elinor Ostrom** (1933–2012; Nobel Memorial Prize 2009) challenged Hardin's model:

- **Hardin's Category Error:** Hardin conflated **Open-Access Regimes** (*res nullius*, unowned and unregulated) with **Common Property Regimes** (*res communes*, communally managed by groups with defined membership and rules).
- **Empirical Evidence:** Ostrom documented long-enduring, community-governed CPRs that functioned sustainably for centuries without state coercion or private markets: Swiss alpine pastures, Spanish **Huerta** canals, and South Indian irrigation tanks (**Eris**).

2.5 Ostrom's 8 Design Principles for Sustainable CPRs

1. **Clearly Defined Boundaries:** Explicit boundaries of the resource and authorized user rights.
2. **Congruence with Local Conditions:** Extraction rules match local ecological realities.
3. **Collective-Choice Arrangements:** Resource users participate in making and modifying operational rules.

4. **Accountable Monitoring:** Monitors audit resource conditions and are accountable to the users.
5. **Graduated Sanctions:** Sanctions start with low warnings and escalate with repeat violations.
6. **Low-Cost Conflict Resolution:** Rapid, local, accessible mechanisms for dispute resolution.
7. **Minimal Recognition of Rights:** External government authorities respect community self-governance.
8. **Nested Enterprises:** Complex, multi-tiered CPRs are organized into polycentric, nested governance layers.

2.6 Game Theory of CPRs & Traditional Indian Community Institutions

- **The Prisoner's Dilemma of the Commons:** When resource appropriators make decisions in isolation without institutional communication, the **Nash Equilibrium** results in mutual defection (over-extraction), leading to sub-optimal collective ruin. Ostrom proved that repeated communication, monitoring, and shared social norms transform the game from a competitive dilemma into an iterated cooperative equilibrium.
- **Traditional Indian CPR Institutions:**
  - **Kuhl Irrigation Networks (Himachal Pradesh):** Gravity-fed community irrigation channels managed by hereditary water masters (**Kohli**), equitably rationing glacial meltwater across terraces for centuries.
  - **Johads of Alwar (Rajasthan):** Traditional earthen check dams revived by **Rajendra Singh** (Tarun Bharat Sangh, "Waterman of India"), capturing monsoon runoff to recharge dead aquifers and revive seasonal rivers (**Arvari River**).
  - **Sacred Groves (Devarakadus / Orans):** Community-protected forest remnants preserved through cultural taboo, acting as critical refugia for endemic botanical and faunal biodiversity.

3. Urbanization Dynamics, Spatial Morphology & Environmental Stress

3.1 Demographic Drivers: Push-Pull Migration and Spatial Sprawl

- **Push Factors:** Agrarian distress, erratic rainfall, crop failures, soil degradation, and mechanization displacing rural farm labor.
- **Pull Factors:** Industrial/service wages, superior healthcare, higher education, civic infrastructure, and social mobility.
- **Spatial Morphology: Urban Sprawl** is low-density, automobile-dependent outward expansion that inflates per capita utility costs and fragments peri-urban agricultural ecosystems. **Ribbon development** along highway corridors creates severe traffic bottlenecks and roadside dumping.

3.2 The Urban Heat Island (UHI) Effect

The **Urban Heat Island (UHI)** is a microclimatic condition where urban centers exhibit higher ambient temperatures than surrounding rural hinterlands (nocturnal delta $\Delta T = 2°C$ to $8°C$):

1. **Thermal Mass & Low Albedo:** Dark asphalt (albedo 0.05–0.10) and concrete (albedo 0.15–0.25) absorb shortwave solar flux during the day and re-radiate sensible infrared heat at night.
2. **Urban Canyon Geometry:** High aspect ratio ($\frac{H}{W}$) buildings restrict the sky view factor (SVF), trapping multiple radiant reflections between vertical walls.
3. **Suppression of Evapotranspiration:** Replacing vegetated soil with impervious concrete converts solar energy into **sensible heat** rather than **latent heat** of evaporation.

4. **Anthropogenic Heat Injection:** Continuous thermal discharge from vehicle engines, air conditioning condenser units, and industrial exhausts.

- **The Bowen Ratio (β):** The ratio of sensible heat flux (H) to latent heat flux (λE):

$$\beta = \frac{H}{\lambda E}$$

In natural wetlands and moist forests with abundant transpiration, $\beta = 0.1 - 0.3$ (most energy drives cooling evaporation). In dense, paved urban cores with zero soil moisture, β surges to **2.0 to 5.0**, ensuring almost all incoming solar flux heats the air.

- **Sky View Factor (SVF):** Dimensionless metric (0 to 1) representing the fraction of open sky visible from a ground point. In deep urban street canyons, low SVF (< 0.3) prevents outgoing longwave thermal radiation from escaping into space, generating persistent nighttime heat traps.

3.3 Urban Hydrology & Runoff Coefficient

Sealing soil with concrete fundamentally alters catchment hydrology:

- **Runoff Coefficient (C):** Fraction of rainfall converted to surface runoff. Natural forest: $C = 0.05 - 0.20$ (80–90% infiltrates); Dense urban core: $C = 0.70 - 0.95$ ($< 15\%$ infiltrates).
- **Hydrograph Impacts:** Stormwater rushes rapidly over concrete into gutters, shortening time of concentration (T_c), inflating peak flood discharge by 200–500%, and precipitating urban flash floods. Meanwhile, baseflow and aquifer recharge collapse, drying urban streams during dry seasons.

3.4 Slums, Environmental Inequity & Civic Infrastructure Deficits

Informal settlements (**slums**) experience severe deficits in basic civic infrastructure:

- **Potable Water & Sanitation Deficit:** Lack of municipal piped water forces reliance on exploitative private tanker mafias. Absence of underground sewerage leads to open drain contamination, causing high child morbidity from cholera and typhoid.
- **Hazard Exposure:** Slums are disproportionately located on vulnerable marginal lands: steep landslide-prone slopes, flood-prone drainage canals, or next to industrial toxic waste dumps.

4. Sustainable Urban Planning, Green Infrastructure & Building Codes

4.1 Transit-Oriented Development (TOD) & Active Mobility

- **TOD Principles:** High-density, mixed-use zoning (residential, commercial, retail) centered within a **400 to 800 meter** walking radius (5–10 min walk) of high-capacity mass transit hubs (Metro, commuter rail, BRTS).
- **Non-Motorized Transport (NMT):** Prioritizing continuous bicycle corridors, shaded sidewalks, and pedestrianized streets to solve the first-and-last-mile connectivity challenge and suppress vehicle kilometers traveled (VKT).

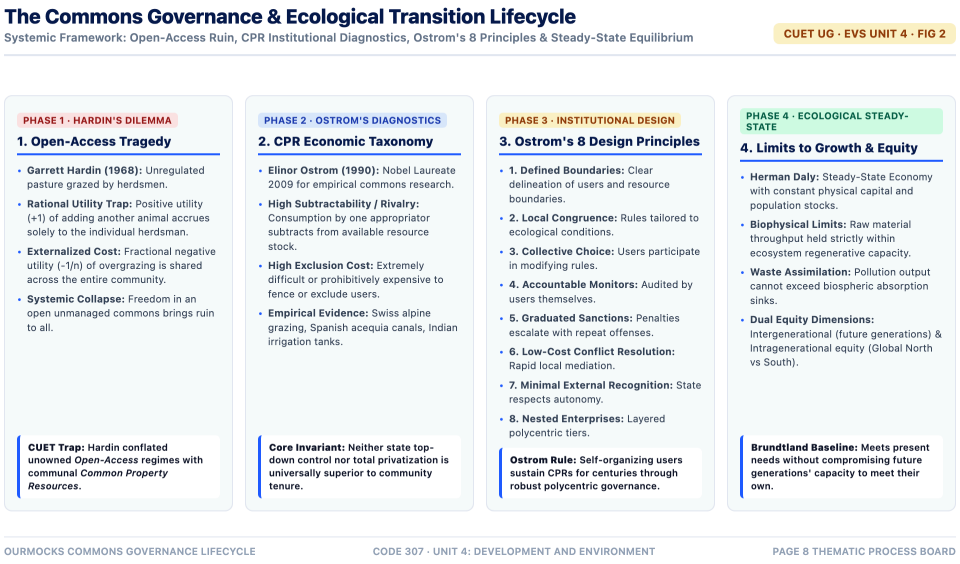


Figure 2: The Commons Governance Lifecycle: Hardin's Open-Access Dilemma, CPR Institutional Diagnostics, Ostrom's 8 Principles, and Steady-State Equilibrium (CUET Unit 4).

4.2 Nature-Based Solutions: Sponge Cities & SuDS

- **Sponge Cities (Kongjian Yu):** Urban infrastructure engineered to absorb, store, filter, and release stormwater naturally using permeable pavements, sunken green swales, rain gardens, and wetland restoration.
- **Sustainable Drainage Systems (SuDS):** Manages surface runoff at its source to mitigate flood risk, purify water, and enhance urban biodiversity.
- **Rooftop Rainwater Harvesting (RWH):** Mandated in Indian municipal building bye-laws to collect rooftop runoff for direct reuse or deep aquifer recharge via desilting percolation pits.

4.3 Green Building Certification Systems

1. **GRIHA (Green Rating for Integrated Habitat Assessment):** India's national rating system, developed by **TERI** and adopted by the **MNRE** in 2007. Evaluates lifecycle building performance across 34 criteria (100-point scale), awarding 1-Star to 5-Star ratings.
2. **LEED:** Developed by the **U.S. Green Building Council (USGBC)** in 1998; tiers: Certified, Silver, Gold, Platinum.
3. **IGBC:** Established by CII in 2001; provides rating systems tailored to Indian climatic conditions.
4. **ECBC (Energy Conservation Building Code):** Launched in 2007 by the **Bureau of Energy Efficiency (BEE)** under the Energy Conservation Act, 2001. Mandates minimum energy efficiency standards for commercial buildings with connected load ≥ 100 kW or contract demand ≥ 120 kVA.

5. Carrying Capacity, Planetary Boundaries & Eco-Equity

5.1 Biophysical Carrying Capacity (K)

Carrying capacity represents the maximum population an ecosystem can sustain indefinitely without degrading its resource base:

1. **Supportive Capacity:** The capacity of ecological systems to replenish essential input resources (freshwater, arable soil, biomass, minerals).

2. **Assimilative Capacity:** The maximum rate at which environmental media (air, water, soil) can absorb, dilute, and neutralize waste without permanent ecological collapse.

5.2 The IPAT Equation (Ehrlich & Holdren, 1971)

$$I = P \times A \times T$$

- **I (Impact):** Total environmental pressure (emissions, resource depletion).
- **P (Population):** Total human population size.
- **A (Affluence):** Per capita consumption level (Consumption / Person).
- **T (Technology):** Environmental damage per unit of consumption (Impact / GDP).
- **Insight:** High-affluence societies with resource-intensive technologies exert vastly higher environmental impacts per capita than larger subsistence populations.

5.3 Rockström's 9 Planetary Boundaries Framework

Formulated in 2009 by **Johan Rockström** and **Will Steffen** (Stockholm Resilience Centre), establishing 9 biophysical thresholds for safe human development:

1. **Climate Change:** Boundary: 350 ppm CO2. **Transgressed** (> 420 ppm).
2. **Biosphere Integrity:** Extinction rate threshold: < 10 E/MSY. **Severely Transgressed** (> 100–1000 E/MSY).
3. **Biogeochemical Flows:** Industrial Nitrogen/Phosphorus cycles. **Severely Transgressed** (massive eutrophication).
4. **Land-System Change:** Retain 75% of global forests. **Transgressed**.
5. **Freshwater Change:** Consumptive blue and green water limits. **Transgressed**.
6. **Novel Entities:** Synthetic chemicals, plastics, PFAS, GMOs. **Transgressed**.
7. **Ocean Acidification:** Aragonite saturation state. **Approaching Boundary**.
8. **Stratospheric Ozone Depletion:** Ozone recovery under Montreal Protocol. **Within Safe Zone**.
9. **Atmospheric Aerosols:** Particulate loading affecting monsoons. **Regionally Transgressed**.

5.4 Intergenerational & Intragenerational Equity

1. **Brundtland Definition (1987):** Sustainable development meets the needs of the present without compromising the ability of future generations to meet their own needs.
2. **Intergenerational Equity:** Planetary trusteeship—the moral duty to bequeath to future unborn generations a biosphere of comparable health, diversity, and resource wealth.
3. **Intragenerational Equity:** Fair distribution of resources and ecological burdens among contemporary societies, addressing the historical emissions deficit between the Global North and Global South (Rio principle of **Common But Differentiated Responsibilities - CBDR**).

5.5 Environmental Justice & Climate Refugees

1. **Environmental Justice Movement:** Sparked in **1982 in Warren County, North Carolina**, protesting against a toxic PCB landfill sited in a poor African American community. Sociologist **Robert Bullard** (**Dumping in Dixie**, 1990) documented systemic **environmental racism** in hazardous facility siting.
2. **Climate Refugees:** Populations displaced by sea-level rise (Tuvalu, Sundarbans), desertification, and floods. They possess **zero legal refugee status** under the 1951 UN Refugee Convention, which strictly covers persecution based on race, religion, nationality, or political affiliation.

5.6 Kate Raworth's "Doughnut Economics" (2017)

British ecological economist **Kate Raworth** synthesized human development and ecological limits into the **Doughnut Framework**:

- **The Social Foundation:** Derived from the internationally agreed minimum standards of the **Sustainable Development Goals (SDGs)**—ensuring no human falls short on essentials: clean water, adequate nutrition, healthcare, education, housing, energy, gender equality, and political voice.
- **The Ecological Ceiling:** Formulated directly from **Rockström's 9 Planetary Boundaries**—ensuring humanity does not overshoot critical Earth system thresholds (climate disruption, ocean acidification, biodiversity collapse, chemical pollution).
- **The Safe and Just Space for Humanity:** The dynamic equilibrium zone ("the sweet spot") between the social foundation and ecological ceiling where humanity can flourish sustainably within biospheric carrying capacity.

6. CUET High-Yield Invariant Master Table

Development Vector	Pioneering Thinker	Core Analytical Model	CUET Examination Core Invariant
Limits to Growth	Club of Rome (1972)	World3 System Dynamics	Overshoot and collapse predicted if exponential industrial growth continues.
Steady-State Economy	Herman Daly (SSE)	Thermodynamic Subsystem	Constant population and physical capital maintained by minimal throughput.
Tragedy of Commons	Garrett Hardin (1968)	Rational Herdsman Parable	Conflated open access with CPRs; advocated coercion or privatization.
Governing Commons	Elinor Ostrom (1990)	8 Design Principles	CPRs sustainably managed by self-organizing user institutions; Nobel 2009.
Ecological Footprint	Rees & Wackernagel	Global Hectares (gha)	Demand on nature vs Biocapacity; Earth Overshoot Day marks deficit.
The IPAT Equation	Ehrlich & Holdren	$I = P \times A \times T$	Impact = Population times Affluence times Technology.
Planetary Boundaries	Johan Rockström (2009)	9 Earth System Thresholds	Biosphere, Climate, Biogeochemical flows severely transgressed.
Urban Heat Island	Urban Microclimate	Delta T = 2°C to 8°C	Low albedo, thermal mass, canyon trapping, lack of evapotranspiration.
Runoff Coefficient	Urban Hydrology	Impervious fraction C	Surges from 10% (forest) to >55% (urban); causes flash floods.
GRIHA Rating	TERI / MNRE (2007)	34 Criteria (100 Pts)	India's indigenous green building code; 1-Star to 5-Star ratings.
Environmental Justice	Robert Bullard (1990)	Warren County (1982)	Protests against toxic PCB landfill siting; environmental racism baseline.

Unit 4: Concept Synthesizer Mindmap

Comprehensive Visual Topology: Growth Limits, Ostrom's Commons, Urban Systems & Environmental Justice

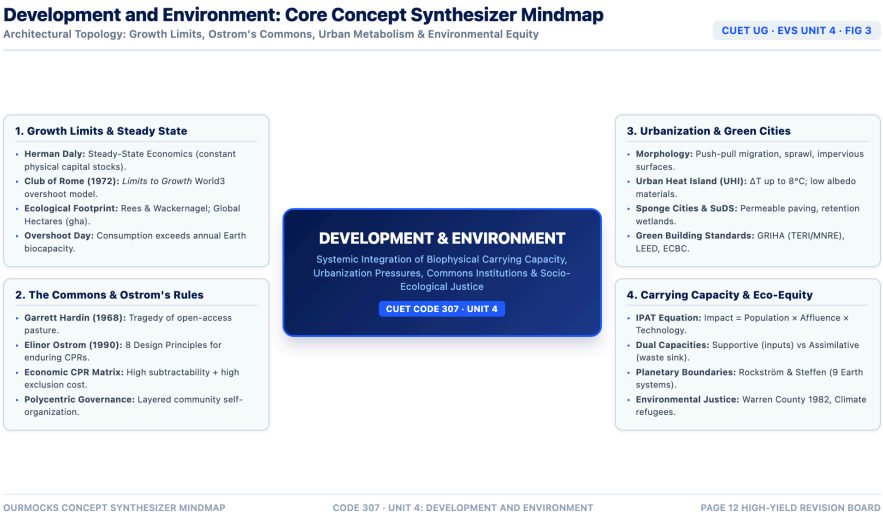


Figure 3: Synthesizer Mindmap: Structural Integration of Growth Limits, Ostrom's CPR Design Principles, Urban Heat Island, and Environmental Equity (CUET Unit 4).

OurMocks

Development and Environment: Quick Reference Cheatsheet

ESSENTIAL THEORETICAL DIMENSIONS & REGULATORY BENCHMARKS

▸ **IPAT Formulation:** $I = P \times A \times T$ (Impact = Population × Affluence × Technology) formulated by Paul Ehrlich & John Holdren. ▸ **Earth Overshoot Day:** $\left(\frac{\text{Planet's Biocapacity}}{\text{Humanity's Ecological Footprint}}\right) \times 365$. Currently marks consumption at approx 1.7 Earths. ▸ **Thermodynamic Subsystem (Herman Daly):** The economy is an **open subsystem** inside a **finite, closed biophysical ecosphere**. SSE requires constant stocks of capital and people maintained at minimal physical throughput. ▸ **CPRs Economic Definition:** High subtractability (rivalry) paired with high cost of exclusion (difficult to fence). ▸ **Ostrom's 8 Principles:** (1) Defined boundaries, (2) Congruent rules, (3) Collective choice, (4) Accountable monitoring, (5) Graduated sanctions, (6) Low-cost conflict resolution, (7) Recognized rights, (8) Nested polycentric tiers. ▸ **Urban Heat Island (UHI):** Ambient temperature delta ($\Delta T = 2°C$ to $8°C$) caused by low albedo ($0.05 - 0.20$), high thermal mass, street canyon radiant trapping, and loss of latent evaporative cooling. ▸ **Runoff Coefficient (C):** Fraction of rainfall converted to surface runoff. Natural forest: $0.10 - 0.20$; Dense urban core: $0.70 - 0.95$. ▸ **GRIHA Rating System:** India's indigenous green building code formulated by TERI/MNRE; 100-point scale across 34 criteria awarding 1-Star to 5-Star ratings. ▸ **Planetary Boundaries (Rockström):** 9 global biophysical systems; Climate change, Biosphere integrity, Land-use change, Freshwater change, Biogeochemical flows, and Novel entities are currently transgressed.

THE SEVEN FATAL CUET TRAPS IN DEVELOPMENT AND ENVIRONMENT

▸ **FATAL TRAP 1 (Hardin vs. Ostrom Core Fallacy):** Candidates frequently assume Garrett Hardin was describing communal property. Hardin described **unregulated open access** with zero communication! Elinor Ostrom proved that true **Common Pool Resources (CPRs)** are governed through community institutions and graduated sanctions. ▸ **FATAL TRAP 2 (GDP Defensive Expenditure Illusion):** Oil spill cleanups, hospital treatments after toxic gas leaks, and rebuilding after floods **increase** GDP! GDP measures market activity, NOT ecological or social welfare. ▸ **FATAL TRAP 3 (CPRs vs. Public Goods Distinction):** Both have high exclusion costs, but CPRs are **rival (high subtractability)**—one herdsman's grazing subtracts grass from others—whereas Public Goods (clean air, national defense) are **non-rival (low subtractability)**! ▸ **FATAL TRAP 4 (GRIHA National Identity):** Do not confuse GRIHA with LEED! LEED was developed by the **U.S. Green Building Council (USGBC)**; **GRIHA** is India's own national rating system developed by TERI and adopted by MNRE! ▸ **FATAL TRAP 5 (UHI Evapotranspiration Invariant):** Candidates confuse sensible heat with latent heat. Urban areas have **higher sensible heat** and **lower latent heat flux** because concrete and asphalt prevent vegetative evapotranspiration! ▸ **FATAL TRAP 6 (Intergenerational vs. Intragenerational Equity):** Intergenerational equity is across **time** (present vs unborn future generations); Intragenerational equity is across **space/societies today** (Global North vs Global South, rich vs poor). ▸ **FATAL TRAP 7 (Climate Refugee Legal Status):** Under the 1951 UN Refugee Convention, climate-displaced individuals **do not** possess legal refugee status! The convention strictly covers persecution based on race, religion, nationality, or political group.

CUET UG Practice Questions: Development and Environment

Q1. According to Herman Daly, what is the thermodynamic relationship between the economy and the biosphere?

(A) An open subsystem embedded within a finite, closed, non-growing biophysical ecosphere
(B) An isolated system capable of circular self-replication independent of biophysical inputs
(C) A closed system capable of infinite growth via continuous technological dematerialization
(D) An expanding system that automatically enlarges Earth's biophysical absorption sinks

Sol. Q1. Daly proved the economy is an open subsystem inside a finite, closed ecosphere, taking in low-entropy matter/energy and releasing high-entropy waste. **Answer: (A)**

Q2. In the economic taxonomy of goods, how are Common-Pool Resources (CPRs) uniquely classified?

(A) Low subtractability (non-rival) and low cost of excluding potential beneficiaries
(B) High subtractability (rival) and high cost of excluding potential beneficiaries
(C) High subtractability (rival) and low cost of excluding potential beneficiaries
(D) Low subtractability (non-rival) and high cost of excluding potential beneficiaries

Sol. Q2. CPRs are characterized by high subtractability (rival consumption) paired with high exclusion cost (difficult to fence or exclude users). **Answer: (B)**

Q3. Which 1972 Club of Rome report used the MIT World3 model to predict overshoot and collapse in the 21st century?

(A) Our Common Future by the Brundtland Commission
(B) Silent Spring by Rachel Carson
(C) The Limits to Growth by Donella Meadows, Dennis Meadows, et al.
(D) The Population Bomb by Paul R. Ehrlich

Sol. Q3. The Limits to Growth (1972) by Donella Meadows et al. used World3 system dynamics to predict overshoot and collapse under business-as-usual. **Answer: (C)**

Q4. What was the pivotal conceptual flaw that Elinor Ostrom identified in Hardin's "Tragedy of the Commons"?

(A) Hardin assumed pasture cattle consume more grass than sheep in communal meadows
(B) Hardin ignored the role of technological genetic modification in increasing grass productivity
(C) Hardin failed to consider that human population growth would naturally stabilize
(D) Hardin conflated unowned open-access regimes with self-governing common property institutions

Sol. Q4. Ostrom proved that Hardin conflated open access (res nullius) with common property (res communes) governed by user communities. **Answer: (D)**

Q5. In the IPAT environmental impact equation ($I = P \times A \times T$), what does the variable 'T' represent?

(A) Technological intensity, representing environmental damage per unit of consumption
(B) Total time elapsed in years since the commencement of rapid fossil industrialization
(C) Thermal heat generation from automotive engines and building HVAC condenser units
(D) Topographical land area converted to residential and commercial built infrastructure

Sol. Q5. In I = P x A x T, 'T' represents technological intensity or environmental inefficiency, measuring impact per unit of economic output. **Answer: (A)**

Q6. Why does the Urban Heat Island (UHI) effect cause nighttime temperatures in cities to remain significantly higher ($\Delta T = 2°C$ to $8°C$)?

(A) Because urban trees transpire three times more water vapor at night than rural crops

(B) Because low-albedo building materials store daytime heat and street canyons trap nocturnal infrared radiation

(C) Because urban winds accelerate at midnight, pulling hot air downward from the stratosphere

(D) Because metropolitan sewage treatment plants discharge boiling water into urban aquifers

Sol. Q6. UHI is driven by low-albedo concrete/asphalt absorbing daytime radiation, street canyons trapping heat, and lack of evaporative cooling. **Answer: (B)**

Q7. How does the replacement of natural soil with impervious surfaces alter urban catchment hydrology during storm events?

(A) It increases soil infiltration and elevates groundwater baseflow during dry summer periods

(B) It lengthens the hydraulic lag time between rainfall peak and stream discharge

(C) It dramatically increases the surface runoff coefficient (C), generating extreme peak flood discharges

(D) It eliminates surface runoff completely by absorbing all rainwater into concrete pores

Sol. Q7. Impervious surfaces surge runoff coefficient from 10% to >55%, accelerating time of concentration and causing flash floods. **Answer: (C)**

Q8. What is GRIHA, and what is its statutory role in India's sustainable urban development framework?

(A) A satellite system operated by ISRO to track illegal deforestation across the Western Ghats

(B) A financial regulatory agency governing carbon credit trading on the National Stock Exchange

(C) A judicial tribunal established under the Environment Protection Act to adjudicate river disputes

(D) India's national green building rating system developed by TERI and adopted by MNRE

Sol. Q8. GRIHA (Green Rating for Integrated Habitat Assessment) is India's national rating tool for green buildings, developed by TERI and adopted by MNRE. **Answer: (D)**

Q9. Which of the following is ONE of Elinor Ostrom's eight structural design principles for sustainable CPR governance?

(A) Clearly defined boundaries of resource users and resource extraction perimeters

(B) Mandatory total privatization of the resource stock to multinational corporate entities

(C) Exclusive reliance on distant federal military forces to police all day-to-day user interactions

(D) Immediate capital punishment for any initial minor infraction of operational harvesting rules

Sol. Q9. Ostrom's first design principle is clearly defined boundaries, ensuring that legitimate users and resource perimeters are demarcated. **Answer: (A)**

Q10. The 1982 protests in Warren County, North Carolina against a toxic PCB landfill catalyzed which movement?

(A) The Deep Ecology movement led by Arne Næss and George Sessions

(B) The Environmental Justice movement against environmental racism documented by Robert Bullard

(C) The Chipko movement for Himalayan forest protection led by Sunderlal Bahuguna

(D) The Club of Rome's initiative for computer-modeled planetary boundary analysis

Sol. Q10. Warren County protests in 1982 against a toxic PCB landfill catalyzed the Environmental Justice movement against environmental racism. **Answer: (B)**

Q11. What is the critical distinction between 'Intergenerational Equity' and 'Intragenerational Equity'?

(A) Intergenerational applies to urban planning, while intragenerational applies to wildlife reserves

(B) Intergenerational concerns gender balance, while intragenerational concerns generational age gaps

(C) Intergenerational concerns justice across time (future generations); intragenerational concerns fairness across contemporary societies

(D) Intergenerational focuses on non-renewable minerals; intragenerational focuses on renewable timber

Sol. Q11. Intergenerational equity protects unborn future generations; Intragenerational equity addresses fairness among present human populations (CBDR). **Answer: (C)**

Q12. Under the Planetary Boundaries framework, which Earth system processes are currently classified as SEVERELY TRANSGRESSED?

(A) Stratospheric ozone depletion and ocean thermal tides

(B) Atmospheric argon gas equilibrium and deep tectonic plate subduction

(C) Lunar gravitational pull and surface solar wind bombardment

(D) Biosphere integrity (extinction rate) and Biogeochemical flows (nitrogen and phosphorus)

Sol. Q12. Biosphere integrity (extinction rate) and Biogeochemical flows (nitrogen/phosphorus fertilizers) are severely transgressed beyond safe limits. **Answer: (D)**

Q13. Transit-Oriented Development (TOD) concentrates high-density mixed-use development within what spatial radius of transit hubs?

(A) Within a comfortable 400 to 800 meter walking distance (5 to 10 minutes) of rapid transit hubs

(B) At least 15 to 20 kilometers away from any metropolitan railway or rapid transit corridor

(C) Exclusively along peripheral inter-state expressways with zero pedestrian sidewalk infrastructure

(D) Within subterranean tunnels located strictly 50 meters beneath underground subway tracks

Sol. Q13. TOD concentrates high-density residential and commercial mixed-use zoning within 400 to 800 meters (5-10 min walk) of transit hubs. **Answer: (A)**

Q14. Why do environmentally displaced persons ('climate refugees') lack legal protection under the 1951 UN Refugee Convention?

(A) Because the 1951 Convention explicitly excludes people displaced by industrial manufacturing plants

(B) Because the 1951 Convention defines a refugee strictly on persecution based on race, religion, nationality, or political group

(C) Because international law mandates that all climate-displaced citizens must relocate to Antarctic stations

(D) Because the IPCC charter legally prohibits any citizen from emigrating across national borders during floods

Sol. Q14. The 1951 UN Refugee Convention restricts refugee status to persecution based on race, religion, nationality, or political opinion. **Answer: (B)**

Q15. In the context of regional carrying capacity (K), what does an ecosystem's 'Assimilative Capacity' specifically measure?

(A) The maximum monetary value of timber and metal ores that an industrial economy can extract annually

(B) The maximum number of tourists permitted to enter a national park per calendar season

(C) The maximum rate at which environmental media can absorb and neutralize waste without permanent ecological collapse

(D) The percentage of solar radiation absorbed by deep ocean currents to drive atmospheric cyclones

Sol. Q15. Assimilative capacity is the threshold rate at which air, water, and soil can absorb and neutralize waste without permanent ecological collapse. **Answer: (C)**